2.10 Properties of Fourier transform

Fourier transform has many important properties. Apart from giving simple solutions of complicated Fourier transform, these properties also help in finding the effect of various time domain operations on frequency domain.

2.10.1. Linearity property

If $g_1(t) \Leftrightarrow G_1(\omega)$ and $g_2(t) \Leftrightarrow G_2(\omega)$

then $a_1g_1(t) + a_2g_2(t) \Leftrightarrow a_1G_1(\omega) + a_2G_2(\omega)$

where a_1 and a_2 are constants

Proof:

$$F\left[a_1g_1(t)+a_2g_2(t)\right] = \int_{-\infty}^{\infty}\left[a_1g_1(t)+a_2g_2(t)\right]e^{-j\omega t}dt$$

$$= \int_{-\infty}^{\infty} a_1g_1(t)e^{-j\omega t}dt + \int_{-\infty}^{\infty} a_2g_2(t)e^{-j\omega t}dt$$

$$= a_1G_1(\omega)+a_2G_2(\omega)$$

2.10.2. Symmetry property

If $g(t) \Leftrightarrow G(\omega)$, then $G(t) \Leftrightarrow 2\pi g(-\omega)$

Proof:

$$g(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} G(\omega)e^{j\omega t}d\omega$$

$$2\pi g(-t) = \int_{-\infty}^{\infty} G(\omega)e^{-j\omega t}d\omega$$

we can interchange the variable t and ω, i.e. let $t \to \omega$, $\omega \to t$, we have

$$2\pi g(-\omega) = \int_{-\infty}^{\infty} G(t)e^{-j\omega t}dt$$

$$\therefore G(t) \Leftrightarrow 2\pi g(-\omega)$$

2.10.3. Time scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|}G\left(\frac{\omega}{a}\right)$$

Proof:

$$F[g(at)] = \int_{-\infty}^{\infty} g(at)e^{-j\omega t}dt$$

let $x = at$, then $dt = dx/a$,

case 1: when $a > 0$,

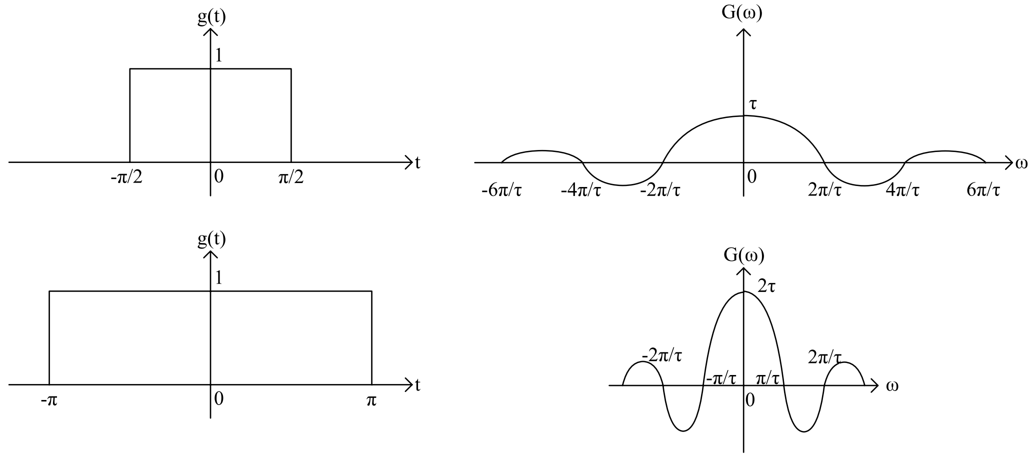
$$F[g(at)] = \frac{1}{a}\int_{-\infty}^{\infty} g(x)e^{-j\omega x/a}dx = \frac{1}{a}G\left(\frac{\omega}{a}\right)$$

case 2: when $a < 0$, then $t \to \infty$ leads to $x \to -\infty$,

$$F[g(at)] = \frac{1}{a}\int_{\infty}^{-\infty} g(x)e^{-j\omega x/a}dx = -\frac{1}{a}\int_{-\infty}^{\infty} g(x)e^{-j\omega x/a}dx = -\frac{1}{a}G\left(\frac{\omega}{a}\right)$$

Combined, the two cases are expressed as,

$$g(at) \Leftrightarrow \frac{1}{|a|}G\left(\frac{\omega}{a}\right)$$



Important Observation:

> Time domain ***compression*** of a signal results in spectral ***expansion***
> Time domain ***expansion*** of a signal results in spectral ***compression***

2.10.4. Time shifting property

$$g(t-t_0) \Leftrightarrow G(\omega)e^{-j\omega t_0}$$

Proof:

$$F[g(t-t_0)] = \int_{-\infty}^{\infty} g(t-t_0)e^{-j\omega t}dt$$

If we let t – t_0 = x so that dt = dx, then

$$F[g(t-t_0)] = \int_{-\infty}^{\infty} g(x)e^{-j\omega(x+t_0)}dx = e^{-j\omega t_0}\int_{-\infty}^{\infty} g(x)e^{-j\omega x}dx = G(\omega)e^{-j\omega t_0}$$

2.10.5. Frequency shifting property

$$g(t)e^{j\omega_0 t} \Leftrightarrow G(\omega-\omega_0)$$

Proof:

$$F[g(t)e^{j\omega_0 t}] = \int_{-\infty}^{\infty} g(t)e^{-j\omega t}e^{j\omega_0 t}dt = \int_{-\infty}^{\infty} g(t)e^{-j(\omega-\omega_0)t}dt = G(\omega-\omega_0)$$

According to this property, multiplication of a function g(t) by exp(jω_0t) is equivalent to shifting its Fourier transform in the positive direction by an amount ω_0, i.e., spectrum F(ω) is translated by an amount ω_0. Therefore, this theorem is also known as **Frequency translation theorem**.

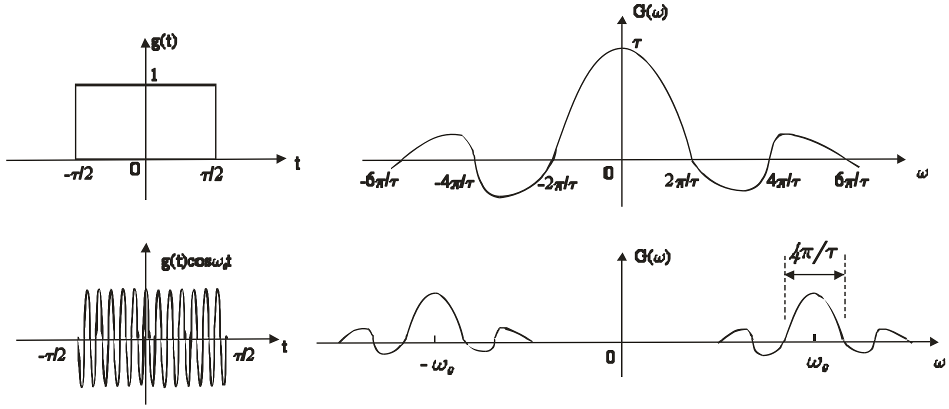
Translation of a spectrum is a very important phenomenon in communication system. This translation helps in achieving the process known as *modulation*. This process can be performed by multiplying the known signal g(t) by a sinusoidal signal. This is because a sinusoidal function can be express as a sum of exponentials as follows

$$g(t)\cos\omega_0 t = \frac{1}{2}[g(t)e^{j\omega_0 t} + g(t)e^{-j\omega_0 t}]$$

Therefore,

$$g(t)\cos\omega_0 t \Leftrightarrow \frac{1}{2}[G(\omega-\omega_0)+G(\omega+\omega_0)]$$

The multiplication of a time function with a sinusoidal function translates the whole spectrum G(ω) to $\pm\omega_0$.
This result is also known as **modulation theorem**.



Example 2.15

To illustrate the usefulness of the Fourier transform linearity and time-shift properties, let us consider the evaluation of the Fourier transform of the signal $x(t)$ shown in **Figure 2.32**(a).

First, we observe that $x(t)$ can be expressed as the linear combination

$$x(t)=\frac{1}{2}x_1(t-2.5)+x_2(t-2.5)$$

where the signals $x_1(t)$ and $x_2(t)$ are the rectangular pulse signals shown in Figure 1.13(b) and (c). Then, using the linearity and time-shifting property, we obtain

$$X_1(\omega)=\frac{2\sin(\omega/2)}{\omega} \text{ and } X_2(\omega)=\frac{2\sin(3\omega/2)}{\omega}.$$

Finally, using the linearity and time-shift properties of the Fourier transform yields

$$X(\omega)=e^{-j5\omega/2}\left\{\frac{\sin(\omega/2)+2\sin(3\omega/2)}{\omega}\right\}.$$

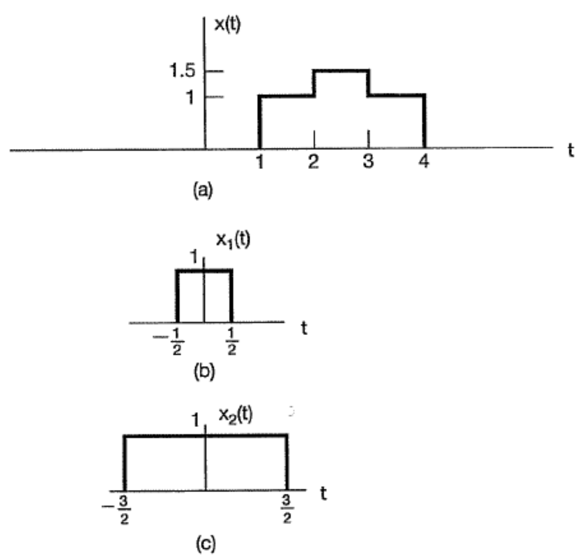


Figure 2.32. Decomposing a signal into the linear combination of two simpler signals. (a) The signal $x(t)$ for Example 2.15; (b) and (c) the two component signals used to represent $x(t)$.

2.10.6. Duality property and scaling property

Example 2.16

Consider the signal $x(t)$ whose Fourier transform is

$$X(\omega)=\begin{cases}1, & |\omega|\le W\\ 0, & |\omega|>W\end{cases}. \tag{2.29}$$

This transform is illustrated in Figure 2.33(a). The signal is then

$$x(t)=\frac{1}{2\pi}\int_{-W}^{W}e^{j\omega t}d\omega=\frac{\sin Wt}{\pi t}=\frac{W}{\pi}\operatorname{sinc}\left(\frac{Wt}{\pi}\right), \tag{2.30}$$

which is depicted in Figure 2.33 (b).

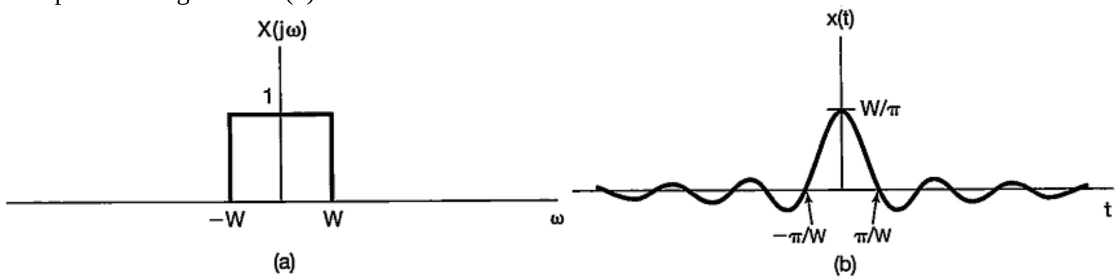


Figure 2.33. Fourier transform pair of Example 2.16:
(a) Fourier transform for Example 2.16 and (b) the corresponding time function.

Comparing **Figure 2.33** and the example from section 2.9.1, we see an interesting relationship. In each case, the Fourier transform pair consists of a function of the form $(\sin a\theta)/(b\theta)$ and a rectangular pulse. However, in section 2.9.1, it is the *signal* $x(t)$ that is a pulse, while in Example 2.16, it is the *transform* $X(j\omega)$. The special relationship that is apparent here is a direct consequence of the *duality property* for Fourier transforms.

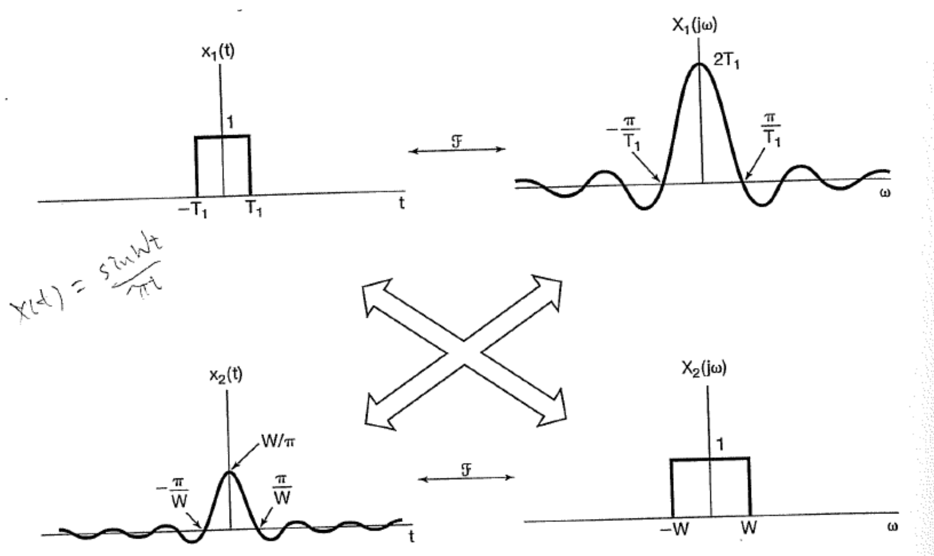


Figure 2.34. Relationship between the Fourier transform pairs to demonstrate the duality property.

Finally, we can gain insight into one other property of the Fourier transform by examining **Figure 2.33**, which we have redrawn as **Figure 2.35** for several different values of W. From this figure, we see that as W increases, $X(\omega)$ becomes broader, while the main peak of $x(t)$ at $t = 0$ becomes higher and the width of the first lobe of this signal (i.e., the part of the signal for $|t| < \pi/W$) becomes narrower. In fact, in the limit as $W \to \infty$, $X(\omega) = 1$ for all ω, and consequently, from Example 2.16, we see that $x(t)$ in eq. (2.30) converges to an impulse as $W \to \infty$. The behavior depicted in **Figure 2.35** is an example of the inverse relationship that exists between the time and frequency domains, or the scaling property of the Fourier transform.

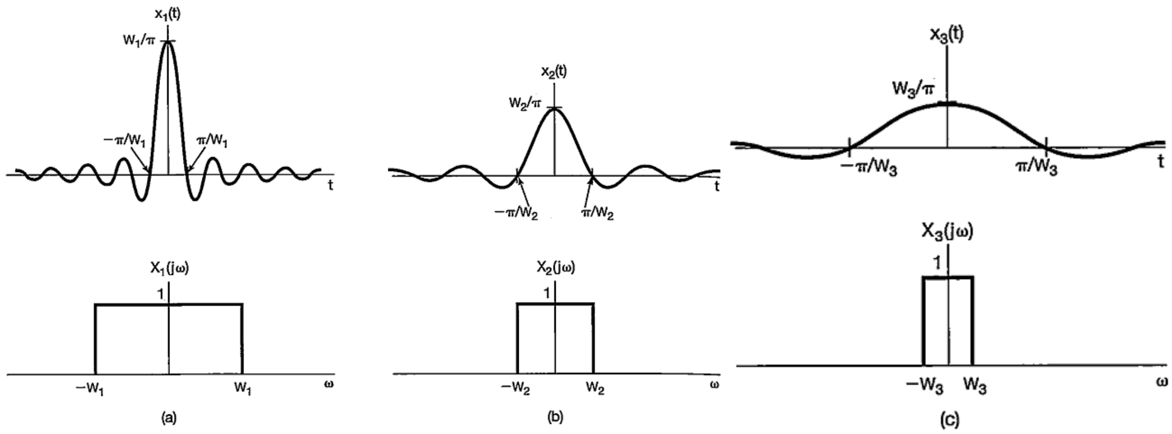


Figure 2.35. Fourier transform pair of **Figure 2.33** for several different values of *W*.

This is a particularly important property, as it replaces the operation of differentiation in the time domain with that of multiplication by $j\omega$ in the frequency domain. We will find the substitution to be extremely useful in our discussion on the use of Fourier transforms for the analysis of LTI systems described by differential equations.

2.10.7. Parseval's Relation

If $x(t)$ and $X(\omega)$ are a Fourier transform pair, then

$$\int_{-\infty}^{+\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(\omega)|^2 \, d\omega. \tag{2.31}$$

This expression, referred to as Parseval's relation, follows from direct application of the Fourier transform. Specifically,

$$\int_{-\infty}^{+\infty} |x(t)|^2 \, dt = \int_{-\infty}^{+\infty} x(t) x^*(t) dt = \int_{-\infty}^{+\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{+\infty} X^*(\omega) e^{-j\omega t} d\omega \right] dt.$$

Reversing the order of integration gives

$$\int_{-\infty}^{+\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X^*(\omega) \left[\int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \right] d\omega.$$

The bracketed term is simply the Fourier transform of $x(t)$; thus,

$$\int_{-\infty}^{+\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(\omega)|^2 \, d\omega.$$

The term on the left-hand side of eq. (2.35) is the total energy in the signal $x(t)$. Parseval's relation says that this total energy may be determined either by computing the energy per unit time ($|x(t)|^2$) and integrating over all time or by computing the energy per unit frequency ($|X(\omega)|^2/2\pi$) and integrating over all frequencies. For this reason, $|X(\omega)|^2$ is often referred to as the *energy-density spectrum* of the signal $x(t)$.

Example 2.17

For each of the Fourier transforms shown in **Figure 2.36**, we wish to evaluate the following time-domain expressions:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt.$$

To evaluate E in the frequency domain, we may use Parseval's relation. That is,

$$E = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(\omega)|^2 \, d\omega \tag{2.32}$$

which evaluates to 10/9 for **Figure 2.36** (a) and to 2 for **Figure 2.36** (b).

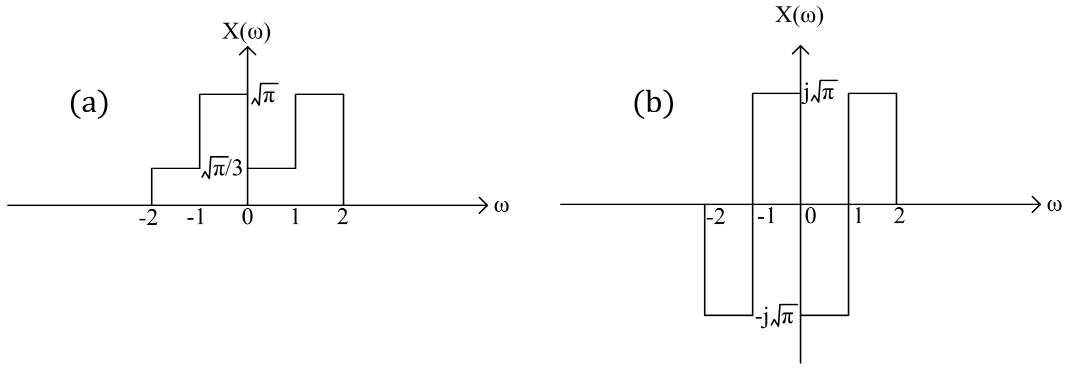


Figure 2.36. The Fourier transforms considered in Example 2.17.

2.10.8. Energy Spectral Density (ESD)

Equation (2.36) can be interpreted to mean that the energy of a signal $g(t)$ is the result of energies contributed by all the spectral components of the signal $g(t)$. The contribution of a spectral component of frequency ω is proportional to $|G(\omega)|^2$. We can interpret $|G(\omega)|^2$ as the energy per unit bandwidth (in hertz) of the spectral components of $g(t)$ centered at frequency ω. In other words, $|G(\omega)|^2$ is the energy spectral density (per unit bandwidth in hertz) of $g(t)$. Actually the energy contributed per unit bandwidth is $2|G(\omega)|^2$ because both the positive and the negative frequency components combine to form the components in the band Δf· However, for the sake of convenience we consider the positive and negative frequency components being independent. Some authors *do* define $2|G(\omega)|^2$ as the energy spectral density. The **energy spectral density (ESD)** $\Psi_g(t)$ is thus defined as

$$\Psi_g(\omega) = \left|G(\omega)\right|^2 \tag{2.33}$$

From the results in Example 2.13, the ESD of the signal $g(t) = e^{-at}u(t)$ is

$$\Psi_g(\omega) = \left|G(\omega)\right|^2 = \frac{1}{\omega^2 + a^2}$$

2.10.9. Bandwidth of a Signal

The spectra of most signals extend to infinity. However, because the energy of a practical signal is finite, the signal spectrum magnitude must approach 0 as $\omega \to \infty$. Most of the signal energy is contained within a certain band of B Hz, and the energy content of the components of frequencies greater than B Hz is negligible. We call B the **bandwidth** of the signal. The criterion for selecting B depends on the error tolerance in a particular application. We may, for instance, select B to be that band which contains 95% of the signal energy. This figure may be higher or lower than 95%, depending on the precision needed. Using such a criterion, we can determine the bandwidth of a signal. Suppression of all the spectral components of $g(t)$ beyond the bandwidth results in a signal $\hat{g}(t)$, which is a close approximation of $g(t)$. If we use the 95% criterion for the essential bandwidth, the energy of the error (the difference) $g(t) - \hat{g}(t)$ is 5% of E_g. The following example demonstrates the bandwidth estimation procedure.

Example 2.18

Estimate the bandwidth W rad/s of the signal $g(t)=e^{-at}u(t)$ if the bandwidth is defined as the range of frequencies that contains 95% of the signal energy.

In this case,

$$G(\omega) = \frac{1}{j\omega + a}$$

and the ESD is

$$\left|G(\omega)\right|^2 = \frac{1}{\omega^2 + a^2}$$

This ESD is shown in Figure 2.37. Let W rad/s be the bandwidth, which contains 95% of the total signal energy E_g. In this case

$$0.95E_g = 0.95 \cdot \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{d\omega}{\omega^2+a^2} = \frac{1}{2\pi}\int_{-W}^{W}\frac{d\omega}{\omega^2+a^2}$$

$$0.95 \cdot \frac{1}{2\pi a}\tan^{-1}\frac{\omega}{a}\Big|_{-\infty}^{\infty} = \frac{1}{2\pi a}\tan^{-1}\frac{\omega}{a}\Big|_{-W}^{W}$$

$$\frac{0.95}{2a} = \frac{1}{\pi a}\tan^{-1}\frac{W}{a}$$

$$W = a\tan\left(\frac{0.95\pi}{2}\right) = 12.706a \text{ rad/s.}$$

This means that the spectral components of $g(t)$ in the band from 0 (dc) to 12.706a rad/s (2.02 Hz) contribute 95% of the total signal energy; all the remaining spectral components (in the band from 12.706a rad/s to ∞) contribute only 5 % of the signal energy.

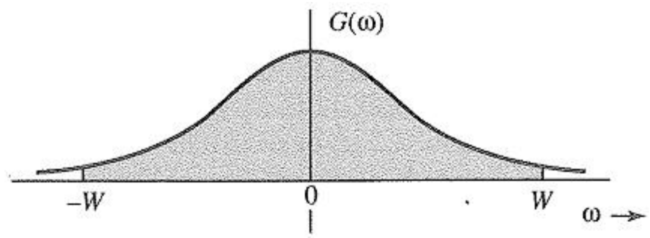
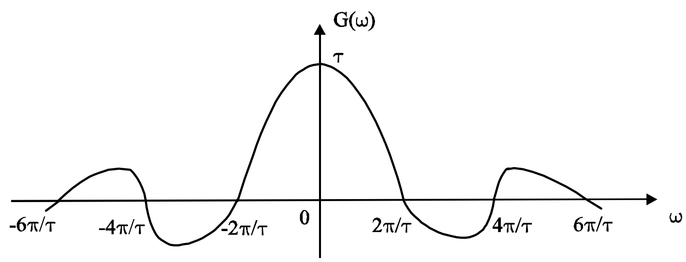
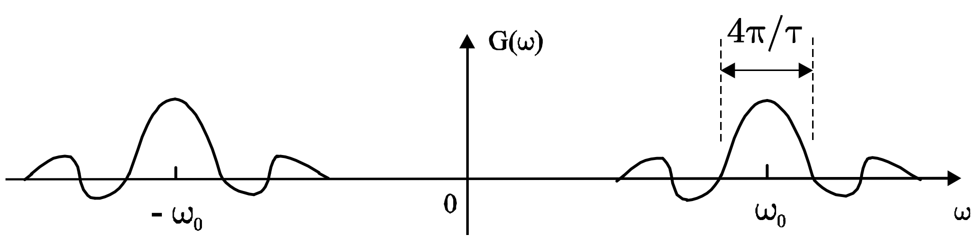


Figure 2.37. Estimating the bandwidth of the signal $g(t)=e^{-at}u(t)$.

Alternatively, the bandwidth of a signal $g(t)$ can be defined as the smallest frequency where G(ω)=0. For example, the following signal has a bandwidth of $2\pi/\tau$.



For signals not centered at 0 frequency , the bandwidth is will be defined as the distance in frequency between two zero-crossings of its spectrum that is closest to the carrier frequency. For example, the following signal has bandwidth $4\pi/\tau$.



2.10.10. Differentiation Property

Let $x(t)$ be a signal with Fourier transform $X(\omega)$. Then, by differentiating both sides of the Fourier transform synthesis equation (2.26), we obtain

$$\frac{dx(t)}{dt} = \frac{1}{2\pi}\int_{-\infty}^{+\infty} j\omega X(\omega)e^{j\omega t}d\omega$$

Therefore,

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$$

2.10.11. Convolution Property

Suppose that $x_1(t) \Leftrightarrow X_1(\omega)$ and $x_2(t) \Leftrightarrow X_2(\omega)$, consider the convolution of two time function $x_1(t)$ and $x_2(t)$ defined by the following integral

$$x(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau .$$

Time convolution theorem

In this case, the Fourier Transform of $x(t)$ is given by

$$F[x(t)] = F[x_1(t) * x_2(t)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau \right] e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} x_1(\tau) \left[\int_{-\infty}^{\infty} x_2(t-\tau) e^{-j\omega(t-\tau)} dt \right] e^{-j\omega\tau} d\tau$$

$$= \int_{-\infty}^{\infty} x_1(\tau) X_2(\omega) e^{-j\omega\tau} d\tau = X_2(\omega) \int_{-\infty}^{\infty} x_1(\tau) e^{-j\omega\tau} d\tau$$

$$= X_1(\omega) X_2(\omega)$$

Therefore, if $x_1(t) \Leftrightarrow X_1(\omega)$ and $x_2(t) \Leftrightarrow X_2(\omega)$

Then $x(t) = x_1(t) * x_2(t) \Leftrightarrow X_1(\omega)X_2(\omega)$

Frequency convolution theorem

If $x_1(t) \Leftrightarrow X_1(\omega)$ and $x_2(t) \Leftrightarrow X_2(\omega)$

Then $x_1(t) x_2(t) \Leftrightarrow \dfrac{1}{2\pi} X_1(\omega) * X_2(\omega)$

The proof is similar to time convolution theorem.

2.11 Signal transmission through a linear and time-invariant system

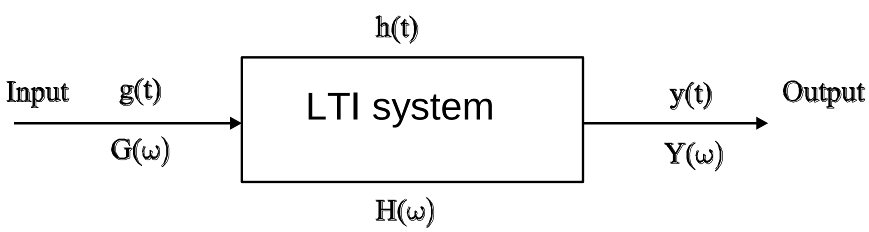


Figure 2.38. .Block diagram of a linear time-invariant (LTI) system.

The time convolution property has important applications in LTI system analysis. Consider an LTI system with input g(t), impulse response h(t) and output y(t). We can write y(t) = g(t) * h(t). Now, if $g(t) \Leftrightarrow G(\omega)$, $h(t) \Leftrightarrow H(\omega)$, $y(t) \Leftrightarrow Y(\omega)$, by convolution theorem

$$Y(\omega) = H(\omega) G(\omega)$$

where *H(ω)* is the **system transfer function or frequency response**. We can write

$$y(t) = h(t) * g(t) \longleftrightarrow Y(\omega) = H(\omega) G(\omega) \tag{2.34}$$

Equation (2.34) is of major importance in signal and system analysis. As expressed in this equation, the Fourier transform maps the convolution of two signals into the product of their Fourier transforms. $H(\omega)$, the Fourier transform of the impulse response, is the frequency response and captures the change in complex amplitude of the Fourier transform of the input at each frequency ω. The frequency response $H(\omega)$ plays as important a role in the analysis of LTI systems as does its inverse transform, the unit impulse response. For one thing, since $h(t)$ completely characterizes an LTI system, then so must $H(\omega)$. In addition, many of the properties of LTI systems can be conveniently interpreted in terms of $H(\omega)$. For example, Using eq. (2.40), we can deduce that that the overall frequency response of the cascade of two LTI systems is simply the product of the individual frequency responses and does not depend on the order in which the systems are cascaded. From this observation, it is then clear that the overall frequency response does not depend on the order of the cascade. Therefore, the impulse response of the cascade of two LTI systems is simply the convolution of the individual impulse responses.

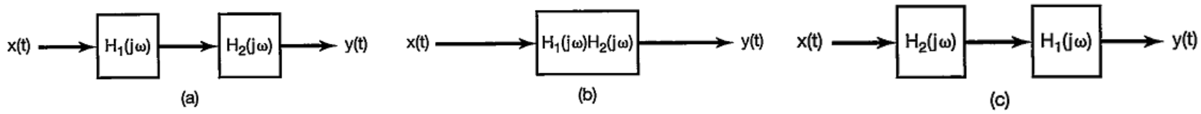


Figure 2.39. Three equivalent LTI systems. Here, each block represents an LTI system with the indicated frequency response.

Example 2.19
Consider a continuous-time LTI system with impulse response

$$h(t) = \delta(t - t_0). \tag{2.35}$$

The frequency response of this system is the Fourier transform of $h(t)$ and is given by

$$H(\omega) = e^{-j\omega t_0}. \tag{2.36}$$

Thus, for any input $x(t)$ with Fourier transform $X(j\omega)$, the Fourier transform of the output is

$$Y(\omega) = H(\omega)X(\omega) = e^{-j\omega t_0}X(\omega). \tag{2.37}$$

This result, in fact, is consistent with the time-shift property. Specifically, a system for which the impulse response is $\delta(t - t_0)$ applies a time shift of t_0 to the input—that is,

$$y(t) = x(t - t_0).$$

Note that, the frequency response of a system that is a pure time shift has unity magnitude at all frequencies (i.e., $\left|e^{-j\omega t_0}\right| = 1$) and has a phase characteristic $-\omega t_0$ that is a linear function of ω.

Example 2.20
As another illustration of the usefulness of the convolution property, let us consider the problem of determining the response of an input signal $x(t)$ that has the form of a sinc function. That is,

$$x(t) = \frac{\sin\omega_i t}{\pi t} = \frac{\omega_i}{\pi}\frac{\sin\omega_i t}{\omega_i t} = \frac{\omega_i}{\pi}\operatorname{sinc}\left(\frac{\omega_i t}{\pi}\right).$$

Let the impulse response of the system is of a similar form, namely

$$h(t) = \frac{\sin\omega_c t}{\pi t} = \frac{\omega_c}{\pi}\frac{\sin\omega_c t}{\omega_c t} = \frac{\omega_c}{\pi}\operatorname{sinc}\left(\frac{\omega_c t}{\pi}\right)$$

with $\omega_c < \omega_i$. The filter output $y(t)$ will therefore be the convolution of two sinc functions, which, as we now show, also turns out to be a sinc function. A particularly convenient way of deriving this result is to first observe that

$$Y(\omega) = X(\omega)H(\omega),$$

where

$$X(\omega) = \begin{cases} 1 & |\omega| \le \omega_i \\ 0 & \text{elsewhere} \end{cases} \qquad \text{and} \qquad H(\omega) = \begin{cases} 1 & |\omega| \le \omega_c \\ 0 & \text{elsewhere} \end{cases}.$$

Therefore

$$Y(\omega) = \begin{cases} 1 & |\omega| \le \omega_c \\ 0 & \text{elsewhere} \end{cases}.$$

Finally, the inverse Fourier transform of $Y(\omega)$ is given by

$$y(t)=\begin{cases}\dfrac{\sin\omega_c t}{\pi t} & \text{if} \quad \omega_c \le \omega_i \\ \dfrac{\sin\omega_i t}{\pi t} & \text{if} \quad \omega_i \le \omega_c\end{cases}.$$

That is, depending upon which of ω_c and ω_i is smaller, the output is equal to either $x(t)$ or $h(t)$.

Example 2.21
Consider the RC-circuit shown below. We can model the circuit as an LTI system with input *g(t)* and output *y(t)* and the circuit itself is the LTI system with **transfer function or frequency response** *H(ω)*.

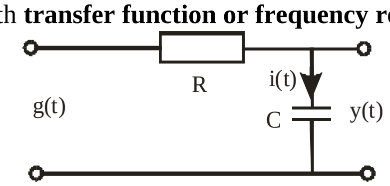


Figure 2.40. An RC circuit modelled as an LTI system with input *g(t)* and output *y(t)*.

Now, how do we obtain *H(ω)?*

From circuit theory, we have $$g(t)=i(t)R+y(t) \quad \text{and} \quad i(t)=C\frac{dy(t)}{dt}.$$

Thus, $$RC\frac{dy(t)}{dt}+y(t)=g(t)$$

Taking the Fourier transform, we have $$j\omega RCY(\omega)+Y(\omega)=G(\omega)$$

Therefore, the transfer function is given by

$$H(\omega)=\frac{Y(\omega)}{G(\omega)}=\frac{1}{1+j\omega RC}.$$

From example 2.13, we know the Fourier transform pair

$$e^{-at}u(t) \Leftrightarrow \frac{1}{a+j\omega}.$$

Therefore, the impulse response of this system is given by $h(t)=\dfrac{1}{RC}e^{-\frac{t}{RC}}u(t)$.

Example 2.22
Consider a LTI system that is characterized by the differential equation

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=3\frac{dx(t)}{dt}+5x(t)$$

Using the differential property, the frequency response is

$$H(\omega)=\frac{3(j\omega)+5}{(j\omega)^2+3(j\omega)+2}. \tag{2.38}$$

To determine the corresponding impulse response, we require the inverse Fourier transform of $H(\omega)$. This can be found using the technique of partial-fraction expansion. As a first step, we factor the denominator of the right-hand side of eq. (2.40) into a product of lower order terms:

$$H(\omega)=\frac{3j\omega+5}{(j\omega+1)(j\omega+2)}. \tag{2.39}$$

Then, using the method of partial-fraction expansion, we find that

$$H(\omega)=\frac{2}{j\omega+1}+\frac{1}{j\omega+2}$$

The inverse transform of each term can be recognized from Example 2.13, with the result

$$h(t)=2e^{-t}u(t)+e^{-2t}u(t).$$